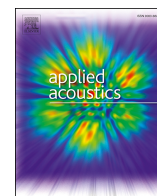




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Quantitative characterization of cavitation features in acoustic emission signals using wavelet packet transforms and k-means clustering

Max Au-Yeung, Reza Haqshenas*

Department of Mechanical Engineering, University College London, Roberts Engineering Building, Torrington Place, London, WC1E 7JE, United Kingdom

HIGHLIGHTS

- Wavelet packet decomposition enables multi-resolution cavitation analysis.
- First algorithmic quantification of broadband bandwidth using optical metrics.
- Hybrid SOM-k-means clustering enables robust feature classification and regime isolation.
- Multi-resolution spectral decomposition separates cavitation components.
- Validated on Gilmore-model simulations and experimental ultrasound data.

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ABSTRACT

Acoustic emissions from cavitation bubbles contain complex information about bubble dynamics. Passive cavitation detectors are widely used to measure these emissions. Processing this data enables cavitation monitoring, which has a broad range of biomedical applications. Traditional methods typically rely on spectral features to identify and characterize cavitation qualitatively. However, because of the non-unique relationship between spectral features and cavitation activity, spectral analyses are often insufficient for characterizing cavitation. This paper presents a novel approach that combines multi-resolution time-frequency analysis using the wavelet packet transform with statistical feature extraction, self-organizing maps, and k-means clustering to separate, quantify, and classify cavitation features in acoustic emissions. We introduce metrics adapted from optics to quantify the frequency bandwidth of the broadband component usually associated with transient cavitation. The method is validated using synthetic acoustic emissions generated by solving single-bubble dynamics via the Gilmore model, and further tested on experimental data from biomedical ultrasound applications.

1. Introduction

Acoustic cavitation refers to the formation, growth, and oscillation of bubbles (cavities filled with gas) in a liquid using acoustic waves [1]. These bubbles may oscillate while exposed to acoustic waves (known as stable cavitation) or grow and implode violently within a few cycles of excitation (known as inertial/transient cavitation). Cavitation activity can generate various physical phenomena, including the creation of free radicals, shock waves, shear stresses, micro-jets, and micro-streaming in the insonated liquid [1], which can trigger diverse physicochemical interactions [2–7]. In biomedical ultrasound, these phenomena and effects are utilized to induce controlled biological responses for therapeutic applications [8–10]. Quantitative measurement and characterization of acoustic cavitation are essential for mechanistic studies and process

monitoring, particularly in therapeutic ultrasound, where it is critical to achieve the desired therapeutic effect while avoiding overtreatment and damage to healthy tissue. Passive cavitation detectors (PCDs) are widely used to measure acoustic emissions from cavitation bubbles in both basic and applied research [9,11]. In therapeutic ultrasound, PCDs have been used in a range of applications, including hyperthermia and thermal ablation [12], histotripsy [13], drug and gene delivery [14], and opening of the blood-brain barrier [15,16]. Acoustic emissions exhibit distinct spectral features influenced by sonication parameters and bubble properties (e.g., uncoated or coated bubbles, vapor pressure of the gas core, surface tension). These include: (i) harmonics at nf where f is the fundamental frequency and n is a positive integer; (ii) sub-harmonics at nf/m where $m > n$ is not divisible by n ; (iii) ultra-harmonics at mf/n ; and (iv)

* Corresponding author.

Email address: r.haqshenas@ucl.ac.uk (R. Haqshenas).

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broadband frequency components. The origin and interpretation of these features remain complex and interdependent, with various hypotheses presented in classical works [17–20] and more recent studies [21–24]. Notably, some of these spectral features are not exclusive to a single cavitation regime. For example, bubbles driven at integer multiples of their resonance frequency may exhibit stable nonlinear oscillations accompanied by sub-harmonics [17,20,22]. Sub-harmonics are also observed in transient cavitation and are linked to prolonged bubble expansion before collapse. In contrast, the broadband spectral component is often associated with inertial collapses in transient cavitation [25]. In bubble clouds, inter-bubble interactions further complicate emissions, producing mixed harmonic content, while shock waves may excite nearby bubbles at their natural resonances, resulting in spectral features unrelated to the excitation frequency [26]. Given the non-uniqueness spectral features of cavitation regimes, binary classification of cavitation based solely on frequency content is unreliable. Theoretical studies of single-bubble dynamics show that stable cavitation tends to produce persistent, frequency-localized components, whereas transient cavitation generates impulsive time-localized components that appear as broadband features. This suggests that combined spectral-temporal analysis is essential to identify time-frequency patterns specific to each regime. Different time-frequency methods were applied to analyze cavitation emissions, such as short time-Fourier transform (STFT) and bilinear time-frequency distributions [25,27–29]. Conventional STFT analysis is limited by its fixed time-frequency resolution, which forces a trade-off between resolving narrow spectral peaks of stable cavitation and capturing the brief, broadband bursts of transient cavitation. Bilinear time-frequency distributions, such as Wigner-Ville and other Cohen's class methods, achieve higher theoretical resolution but suffer from cross-term interference and noise sensitivity in multi-component cavitation signals, complicating quantitative interpretation. In contrast, the discrete wavelet transform (DWT) and its extension, the discrete wavelet packet transform (DWPT), offer adaptive, multi-resolution analysis, using narrow time windows at high frequencies to detect transients and wider windows at low frequencies to resolve periodic content. This makes them better suited for simultaneous detection of stable and transient cavitation signatures. We previously developed an algorithm based on the DWT and statistical analysis to process cavitation emissions [30]. It was observed that the fixed frequency resolution of the DWT limited separation and quantification of closely spaced sub-/ultra-harmonic features, particularly when they occurred within the same wavelet filter bandwidth. This paper presents a new method combining the discrete wavelet packet transform, statistical signal processing, and k-means clustering to quantitatively analyze acoustic emissions. The DWPT enables flexible decomposition of the signal into time-frequency structures with improved resolution, addressing the limitations of the DWT analysis. The statistical properties of each subband are used to classify DWPT nodes and extract relevant features of cavitation. A subset of optimal metrics is selected using self-organizing maps (SOMs), and a new quantitative thresholding scheme is proposed to estimate the energy and bandwidth of broadband components. These new contributions enable the identification and quantitative characterization of both transient and periodic features in emissions. The method is implemented in a Python library, which is available in a GitHub repository [31]. The method is discussed in detail in Section 2. To test and verify the accuracy and efficiency of the method, synthetic acoustic emissions were generated by solving the single bubble dynamics problem using the Gilmore model, presented in Section 2.5. The method was then used to analyze the theoretical acoustic emissions. Finally, the method was used to process experimental PCD data, explained in Section 3.3.

2. Materials and methods

2.1. Discrete wavelet packet transform

The DWPT decomposes a signal into its constituent time-frequency components. In the fast orthogonal DWPT, a bank of conjugate mirror

filters and downsampling (by a factor of 2) are used to decompose a signal $x[n]$ of length $N = 2^{n_0}$ into a cascade of coarser approximations $a_j[n]$ (low-frequency) and finer details $d_j[n]$ (high-frequency) at a scale j [32]

$$\begin{aligned} a_{j+1}[n] &= \sum_{m=0}^{M-1} h[m]a_j[m-2n] \\ d_{j+1}[n] &= \sum_{m=0}^{M-1} g[m]a_j[m-2n] \end{aligned} \quad (1)$$

where $a_0[n] = x[n]$, $h[m]$ and $g[m]$ are the impulse responses of the filters corresponding to the scaling and wavelet basis functions and are related as follows

$$g[m] = (-1)^{1-m}h[1-m] \quad (2)$$

$g[m]$ and $h[m]$ can be viewed as transfer functions of low- and high-pass filters, respectively, with length M . Both approximation coefficients (a_j) and the detail coefficients (d_j) are decomposed at each scale. This results in a full binary tree of subbands. Each node in the binary tree (a wavelet packet) includes a set of coefficients corresponding to a frequency subband and is labeled with a tuple (j, k) where j is the scale number (the depth of the node in the tree) and k is the number of nodes that are on its left at the same scale.

Unlike DWPT with a predefined depth, the proposed algorithm adaptively determines whether a node should be further decomposed based on its statistical characteristics. A node is decomposed when its children contain an imbalance in energy or kurtosis, i.e., one set of coefficients has a significantly higher energy or kurtosis than the other. This indicates that filtering is useful for narrowing down the frequency band of a significant spectral feature. These conditions are implemented by calculating the energy and kurtosis of each child node (see Eqs. (3) and (7)) and testing if the ratio between their energy and kurtosis is greater than a threshold value. Furthermore, a node will not be decomposed if it contains low energy, indicating that the node would contain negligible information upon further decomposition. The energy threshold is given by a user-defined ratio of the total energy of the original signal. As such, the decomposition is governed by three parameters: the energy ratio, k_E , the kurtosis ratio, k_k , and the energy threshold E_t . This new decomposition algorithm produces an asymmetric tree with variable decomposition depth, determined by the transient and harmonic components within each node. Therefore, the decomposition tree follows the underlying time-frequency structure of the acoustic emission, providing high resolution only where additional information is present while avoiding unnecessary decomposition of insignificant components. A flowchart of this method is shown in Fig. 1.

The wavelet packets can be separately reconstructed into signals that combine to form the original signal. This enables the transform to separate periodic and transient components of the original acoustic emission, which tend to have different time-frequency signatures. Once separated, the packets can be classified to indicate the presence of different cavitation features and quantify them.

2.2. Statistical metrics

The DWPT coefficients of the leaf nodes, i.e., the nodes without further decomposition, are quantified using a set of statistical metrics, including kurtosis, crest factor, skewness, entropy, spectral entropy, energy, and logarithm of energy. The choice of these metrics is guided by the spectral and temporal characteristics of the periodic and transient cavitation described in Section 1. This approach allows for consistent characterization of packets of varying lengths, produced by asymmetric DWPT decomposition, using a uniform and compact set of descriptors, which together form the feature matrix for the leaf nodes.

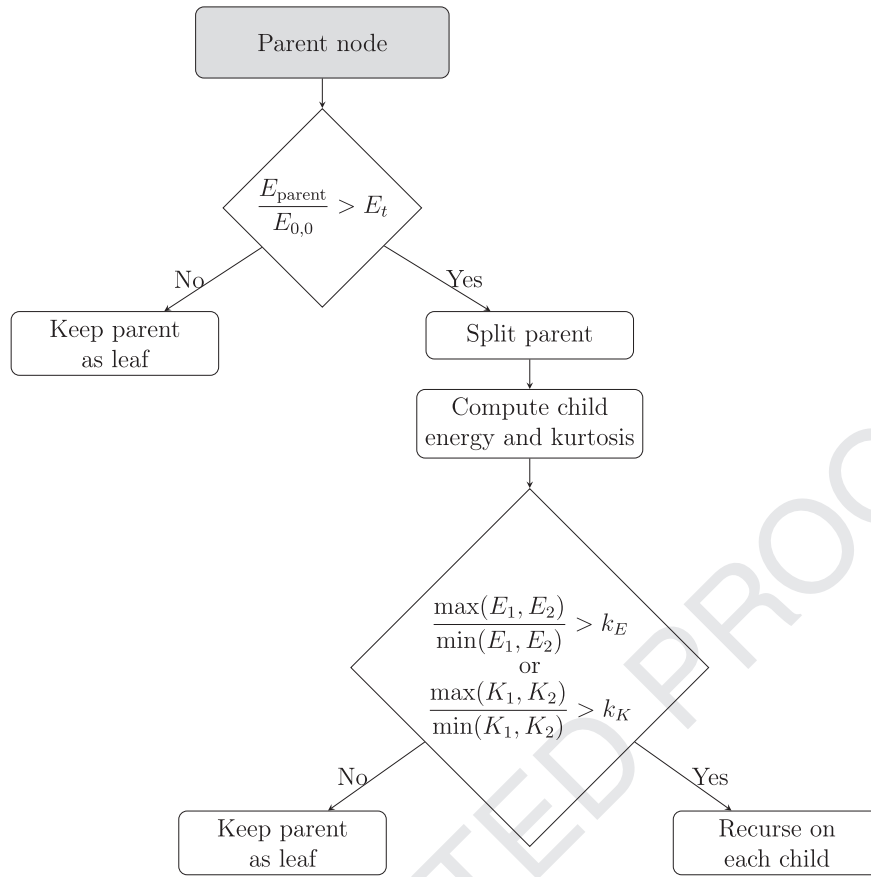


Fig. 1. Adaptive DWPT decomposition: a parent node is first tested against the minimum energy threshold. If the threshold is satisfied, the node is split into two children and the energy and kurtosis ratios are evaluated. The decomposition continues recursively with each child node as a new parent node.

The energy of a packet, which measures how much of the signal is concentrated within a given subband, is calculated as follows:

$$E_{j,k} = \sum_{n=1}^{N_{j,k}} x_{j,k}[n]^2 \quad (3)$$

where $x_{j,k}[n]$ are the coefficients ($a_{j,k}$ and $d_{j,k}$) in a subband (j,k) and $N_{j,k}$ is the number of coefficients in the subband. $E_{0,0}$ is the total energy of the original signal. Note that using orthogonal wavelet bases for DWPT preserves the energy of the signal. Hence, the sum of the energy of all nodes in the final scale equals the energy of the time series signal. The average energy of the coefficients in a node is defined as

$$\bar{E}_{j,k} = \frac{1}{N_{j,k}} E_{j,k} \quad (4)$$

where $E_{j,k}$ is the energy of the coefficients in a subband (j,k) and $N_{j,k}$ is the number of coefficients in the subband.

To better distinguish fundamental, sub- and ultra-harmonics, the log-energy metric is also used, i.e., $LE_{j,k} = \log E_{j,k}$. To measure the energy distribution of a subband—compared to other subbands—Shannon entropy is used, defined as follows:

$$S_{j,k} = - \sum_{n=1}^{N_{j,k}} \frac{x_{j,k}[n]^2}{E_{j,k}} \log \frac{x_{j,k}[n]^2}{E_{j,k}}. \quad (5)$$

Level entropy is a variation of the entropy calculation given by

$$S'_{j,k} = - \sum_{n=1}^{N_{j,k}} \frac{x_{j,k}[n]^2}{E_{0,0}} \log \frac{x_{j,k}[n]^2}{E_{0,0}} \quad (6)$$

The normalized spectral entropy $S(f)_{j,k}$ is a measure of the complexity of the distribution in the spectrum of the subband. This is calculated using Eq. (5) where $x_{j,k}[n]$ is replaced with its power spectral density normalized by its sum. The normalized spectral entropy is in $[0,1]$ with the lower the normalized spectral entropy of a subband, the more deterministic components (information about the fundamental and harmonics) are within the subband. Subbands with flat power spectral density have values close to 1, corresponding to the highest possible entropy which is a characteristic of white noise. The transient components have a broadband frequency signature; hence a higher entropy can be attributed to transient and/or noise components of the signal. To distinguish these features additional statistical metrics are required.

Kurtosis measures the occurrence of outliers in a signal. Therefore, it is considered a metric for detecting transient impulses generated by transient cavitation. The Pearson definition of kurtosis is calculated using the following formula:

$$K_{j,k} = \frac{1}{n\sigma_{j,k}^4} \sum_{n=1}^{N_{j,k}} (x_{j,k}[n] - \bar{x}_{j,k})^4 \quad (7)$$

where $\bar{x}_{j,k}$ and $\sigma_{j,k}$ are the mean and standard deviation of the coefficients of the node (j,k) .

Spectral kurtosis calculates the kurtosis of the normalized spectrum of the coefficients. Deterministic components have spikier spectra resulting in higher spectral kurtosis.

Skewness measures the extent to which a distribution is asymmetrical. It is given by

$$Sk_{j,k} = \frac{1}{n\sigma_{j,k}^3} \sum_{n=1}^{N_{j,k}} (x_{j,k}[n] - \bar{x}_{j,k})^3 \quad (8)$$

The absolute value of the skewness of the coefficients is expected to be larger in magnitude when a signal includes transient components.

The crest factor is the ratio between the peak value and the mean of the signal, indicating how extreme the peaks are in a time series. It is obtained by

$$C_{j,k} = \frac{\max(|x_{j,k}[n]|)^2}{x_{rms}^2} \quad (9)$$

where x_{rms} is the root-mean-square of $x_{j,k}[n]$.

A feature matrix is formed from the statistical metrics calculated for each packet. The next step is to classify these packets using this feature matrix which is discussed in the next section.

2.3. Self-organizing maps and k-means clustering for feature classification

Considering that labeled PCD data is currently unavailable and the non-uniqueness of spectral features in cavitation signals, an unsupervised clustering algorithm is adopted to classify statistical descriptors extracted from the DWPT coefficients. Self-Organizing Maps (SOMs) are employed to explore the structure of the high-dimensional feature space and distinguish between meaningful and redundant statistical metrics for classification. Once this feature subset has been selected, the SOM is no longer used. The selected statistical features extracted from the DWPT leaf nodes are standardized and provided directly as input to the k-means clustering algorithm. SOMs are a type of neural network that project high-dimensional data onto a low-dimensional (2D here) grid while preserving topological relationships [33]. This projection facilitates the detection of subtle patterns and nonlinear dependencies in the data and enables the visual interpretation of correlations between features [34]. Moreover, using SOMs to identify and only utilize a compact set of physically meaningful features enhances the robustness and performance of the classification framework under varying noise and acquisition conditions.

k-means is a commonly used technique for unlabeled classification [35,36]. It partitions the data into k clusters by minimizing the total intra-cluster variance, defined as the sum of the squared Euclidean distances between each data point and its assigned cluster centroid. Since k-means is a non-convex optimization problem, its results depend on the initial positions of the centroids. To mitigate this variability, the clustering is repeated multiple times with different initializations, and the solution with the lowest loss is selected. Additionally, the k-means++ algorithm [37] is used to initialize centroids in a way that favors well-separated starting points, improving convergence and reducing the likelihood of suboptimal clustering due to early centroid merging.

The appropriate number of clusters k is not known a priori and varies depending on the physics of cavitation and the characteristics of the DWPT-derived features. To address this, we use the silhouette score to determine the optimal number of clusters [38–40]. The silhouette score quantifies the quality of clustering by comparing the average intra-cluster distance (a) with the lowest average inter-cluster distance (b) for each point, and is defined as:

$$s = \frac{b - a}{\max(a, b)} \quad (10)$$

where a higher score (close to 1) indicates that the sample is well matched to its own cluster and poorly matched to neighboring clusters. The silhouette score is calculated for k within a specified range to maximize the average silhouette score, providing a robust and interpretable measure of the validity of clustering [41].

2.4. Method overview

The proposed algorithm is summarized in Fig. 2. The PCD data are mean-centered and normalized by their standard deviation. Next, the

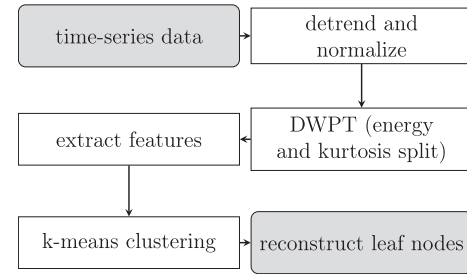


Fig. 2. Flowchart of the proposed method.

DWPT is applied, decomposing the signal into frequency subbands using the energy and kurtosis splitting algorithm (see Section 2.1). The data at each node are trimmed at the edges to remove edge effects by discarding all coefficients that arise from signal padding; then statistical metrics are calculated for each node. These extracted statistical features are then normalized using a standard scaler, mapping them to a standard normal distribution to mitigate scale differences that could bias clustering. These normalized feature vectors are input into the k-means clustering algorithm to identify clusters based on spectral and temporal characteristics. Although the wavelet function must be selected a priori, both the decomposition depth and the number of clusters are determined automatically within the algorithm.

The algorithm is implemented as an open-source library in Python [31] and validated using synthetic acoustic emissions generated by numerically solving the Gilmore equation. Its performance is then tested and verified on experimental PCD data.

2.5. Gilmore model

Gilmore's equation describes the dynamics of a spherical bubble in a compressible liquid undergoing symmetric oscillations due to acoustic excitations. The model has been validated against experimental data and is widely used to study acoustic cavitation, especially in the transient regime [21]. The Gilmore equation is expressed as follows [42]:

$$R\ddot{R}\left(1 - \frac{\dot{R}}{C}\right) + \frac{3}{2}\dot{R}^2\left(1 - \frac{\dot{R}}{3C}\right) = H\left(1 + \frac{\dot{R}}{C}\right) + \frac{R\dot{H}}{C}\left(1 - \frac{\dot{R}}{C}\right) \quad (11)$$

where R is the instantaneous bubble radius, C and H are the speed of sound in the liquid at the bubble wall, and the enthalpy difference in the liquid at the bubble wall (at pressure $P(R)$) and at infinity (P_∞), respectively. P_∞ is the sum of the static reference pressure P_0 and the acoustic pressure $P_{ac}(t)$, i.e., $P_\infty(t) = P_0 + P_{ac}(t)$. Time derivatives are designated by dots. Gilmore's model implicitly assumes that the acoustic pressure is applied instantaneously and uniformly around the bubble without specifying a source location in the liquid. Thus, the liquid density and speed of sound at infinity are taken at the static reference state, denoted by ρ_0 and c_0 , respectively, below. C and H are calculated using the following equations:

$$C = \sqrt{c_0^2 + (n-1)H} \quad (12)$$

$$H = \frac{nA^{1/n}}{(n-1)\rho_0} \left[(P(R) + B)^{(1-1/n)} - (P_\infty(t) + B)^{(1-1/n)} \right] \quad (13)$$

where A , B , and n are constants in the Tait's equation of state which are obtained experimentally. The time derivative of the enthalpy in the liquid at the bubble wall becomes

$$\dot{H} = \frac{A^{1/n}}{\rho_0} \left[\dot{P}(R)(P(R) + B)^{-1/n} - \dot{P}_{ac}(t)(P_\infty(t) + B)^{-1/n} \right]. \quad (14)$$

Note that $\dot{P}_\infty(t) = \dot{P}_{ac}(t)$. $P(R)$ is calculated using a modified Laplace equation

$$P(R) = \left(P_0 + \frac{2\sigma}{R_0} \right) \left(\frac{R_0}{R} \right)^{3\gamma} - \frac{2\sigma}{R} - \frac{4\mu\dot{R}}{R} \quad (15)$$

where R_0 is the initial bubble radius, γ , μ and σ are the adiabatic index (ratio of specific heats), dynamic viscosity, and surface tension, respectively. Finally, taking the time derivative of the above equation, we arrive at

$$\dot{P}(R) = -3\gamma \frac{\dot{R}}{R} \left(P_0 + \frac{2\sigma}{R_0} \right) \left(\frac{R_0}{R} \right)^{3\gamma} + \frac{\dot{R}}{R} \frac{2\sigma}{R} + 4\mu \left(\frac{\dot{R}}{R} \right)^2 - \frac{4\mu\ddot{R}}{R}. \quad (16)$$

Eqs. (12)–(16) are used to numerically solve the Gilmore's equation Eq. (11). The radiated pressure in the liquid is then given by the Vokurka formula [1]

$$\left(\frac{p}{p_{inf}} - 1 \right) \frac{p_{inf}}{\rho} = \frac{R}{r} (\dot{R}R + 2\dot{R}^2) - \frac{\dot{R}^2}{2} \left(\frac{R}{r} \right)^4 \quad (17)$$

where r is the distance from the center of the bubble. This equation assumes that acoustic emissions from an oscillating bubble propagate as outward traveling spherical waves. The Gilmore model is nondimensionalized and solved numerically using a stiff ordinary differential equation solver for different sonication conditions with a sampling rate around 1000 times the excitation frequency. In these simulations, the incident ultrasound pressure is modeled as a sine wave, i.e., $P_{ac}(t) = P_A \sin(2\pi ft)$ where P_A and f are the amplitude and frequency of the ultrasound wave. The implementation is included in the GitHub repository [31]. The implementation and numerical results have been verified against published results for identical parameters. White Gaussian noise (WGN) with the signal-to-noise ratio of 20 dB was added to the calculated pressures to simulate measurement noise.

3. Results

3.1. Feature selection

Many statistical features show potential in classifying bubble dynamics (see Section 2.2). However, given that the k-means algorithm struggles with sparse, high-dimensional data [43], feature selection is required to improve clustering performance.

The feature space was created for acoustic emissions simulated by the Gilmore model. The model was solved for various combinations of ultrasound and bubble parameters, producing 192 signals. The parameters were linearly spaced over the ranges $R_0 \in [1, 7] \mu\text{m}$, $f \in [0.5, 2] \text{ MHz}$, $P_A \in [0.1, 3] \text{ MPa}$. The parameter ranges were selected to span ultrasound conditions commonly encountered in therapeutic ultrasound applications while generating both stable and transient cavitation responses. Varying the initial bubble radius, excitation frequency, and pressure over these ranges produces a diverse set of acoustic emissions containing harmonic, subharmonic, ultra-harmonic, and broadband features, enabling the proposed feature-selection framework to be evaluated across a broad range of physically relevant bubble dynamics. The WGN was added to the signals and they were windowed and normalized as described in Fig. 2 - see Section 3.2 for computational details and parameters.

The SOM is shown in Fig. 3. This consists of a grid of nodes (or cluster centers), each associated with a set of weights, one per feature, that are iteratively adjusted to fit the data. During training, nodes that are close to the data in feature space (i.e., with minimal error relative to nearby data points) adapt their weights to represent the underlying distribution. In the resulting heat maps, each square corresponds to a single node, and the color represents the value of one statistical feature at that node. Separate heat maps are generated for each feature, allowing visual comparison of their distributions. The spatial arrangement of nodes

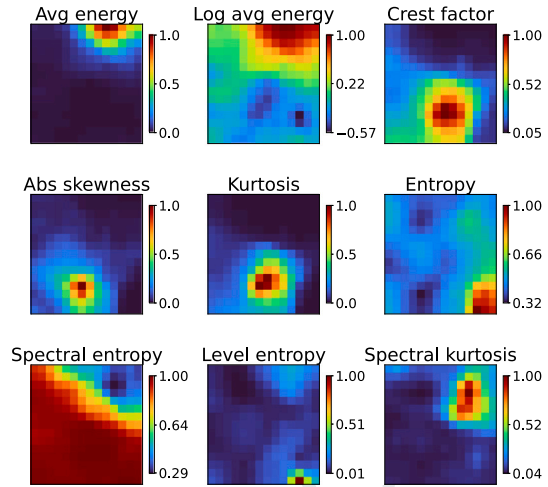


Fig. 3. The SOM is presented, with each feature plotted to illustrate its relationships with other features. The data were generated by solving the Gilmore model with parameters explained in the text. The DWPT was carried out using decomposition levels of 7, with the sym20 wavelet.

is such that those located near each other in the grid are also close in the high-dimensional feature space, making it possible to identify clusters, correlations, and transitions between regimes. Although SOM topology can vary with random initialization, the underlying feature correlations remain consistent across runs. This SOM shows that high energy DWPT nodes tend to have low skewness, kurtosis, and normalized spectral entropy, as expected from the physical interpretations; these correspond to packets containing fundamental frequencies. The normalized spectral entropy and spectral kurtosis are able to find nodes at the other high energy area in the maps. These represent harmonics that are not easily picked out with average energy alone. The nodes with high spectral entropy and low energy indicate signals with sharp transients, while nodes with high normalized spectral entropy but low kurtosis and low energy are indicative of measurement noise.

Feature redundancy is evident in the SOM results: skewness, kurtosis, and crest factor yield similar grids, while normalized spectral entropy and level entropy (normalized by the original signal energy) are also strongly correlated. Based on this observation, the most informative and non-redundant features were identified as log-average energy, kurtosis, and normalized spectral kurtosis. These metrics are used to distinguish periodic, transient, and noise components within the DWPT decomposition. It should be noted that the k-means algorithm clusters DWPT packets according to their statistical descriptors rather than assigning physical cavitation regimes. Depending on the statistical properties of the measurement noise, different threshold values can be defined to distinguish between the cavitation broadband component and the measurement noise. Here, the nodes were thresholded based on the kurtosis value of 3, assuming the measurement noise is white Gaussian. The metrics of the centers of the clusters were used for thresholding.

3.2. Bubble dynamics simulations

In these numerical experiments, the frequency and pulse length were set to $f = 1.1 \text{ MHz}$ and $100/f$, and $P_A = 1 \text{ MPa}$, these values were chosen to match the experimental conditions used for the PCD measurements. We chose water as the surrounding liquid and an air-filled bubble. The physical parameters used for numerical simulations are as follows (at room temperature 293 K and atmospheric pressure $p_0 = 100\text{kPa}$): $\rho_0 = e3 \text{ kg m}^{-3}$, $\sigma = 7.25 \times 10^{-2} \text{ N m}^{-1}$, $\mu = e-3 \text{ Pa s}$, $c_0 = 1482 \text{ m s}^{-1}$, $n = 7$, $A = 3.04 \times 10^2 \text{ MPa}$, $B = 3.039 \times 10^2 \text{ MPa}$ and $\gamma = 1.4$. The initial conditions are $R(0) = R_0$ and $\dot{R}(0) = 0$. The radius of a resonant bubble at the driving frequency (1.1 MHz) is approximately 3.4 μm . Here, three scenarios are simulated in which the initial radii are (i) much smaller

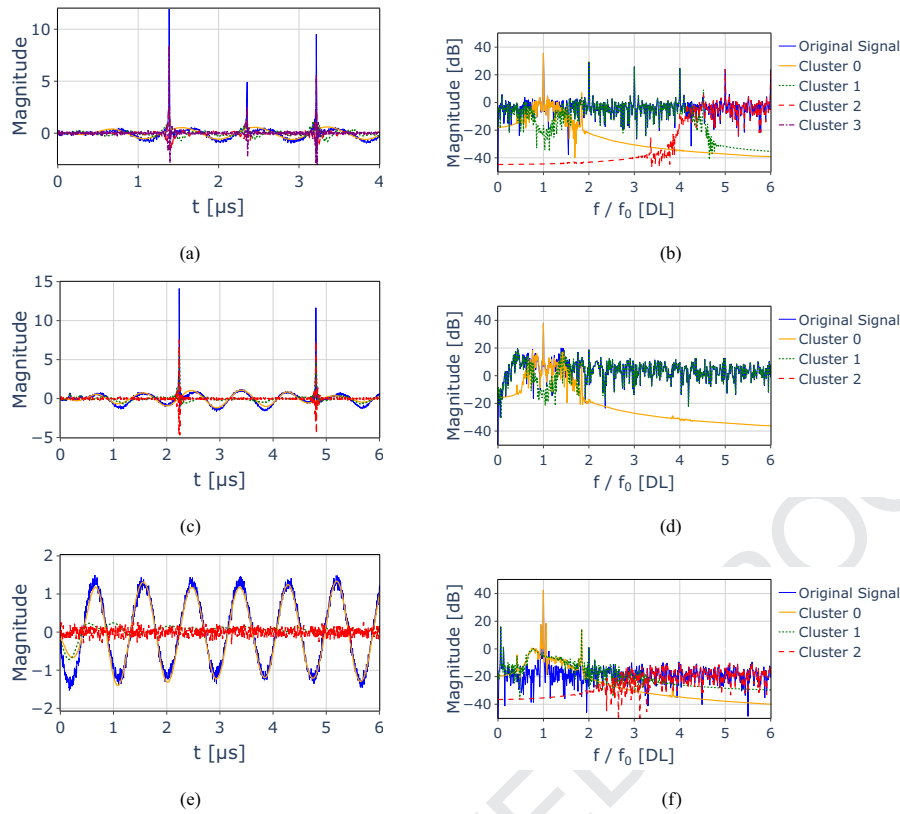


Fig. 4. Reconstructed components of acoustic emissions and their spectra, generated from bubbles with the initial radii of (a,b) 0.5 μm , (c,d) 3.4 μm , and (e,f) 50 μm .

than the resonant bubble; $R_0 = 0.5 \mu\text{m}$, (ii) the resonant bubble radius; $R_0 = 3.4 \mu\text{m}$, and (iii) much larger than the resonant bubble; $R_0 = 50 \mu\text{m}$, resulting in stable and transient cavitation regimes.

The radiated pressures at $r = 100 \mu\text{m}$ and their spectra are shown in Fig. 4.

The bubble with $R_0 = 50 \mu\text{m}$ sustains periodic stable oscillations and radiates cyclic pressure. The main components of the spectrum are the bubble's resonance frequency at $0.06f$ and the fundamental. The transient component has low energy and kurtosis ($K = 3.5$), representing the added WGN (the kurtosis of the WGN is approximately 3). The $R_0 = 0.5 \mu\text{m}$ bubble experiences cyclic transient cavitation, which is realized as impulsive transient components in the emissions. The reconstructed transient component of the acoustic emission manifests harmonics, ultra-harmonics at $(2m+1)f/2$, $m \geq 2$ and broadband spectral components. In this case, the bubble is driven below its resonance, approximately 11.7 MHz (estimated using Minnaert equation); thus, $f/f_r \approx 1/10$. The emissions from the resonant bubble of $R_0 = 3.4 \mu\text{m}$, include the $3f/4$ and $f/2$ sub-harmonics, the fundamental, an ultra-harmonic at about $3f/2$, the second harmonic, and broadband.

To analyze the emissions, the Symlet wavelet function of order 20 (sym20) was chosen in accordance with the physics of cavitation and the impulsive nature of desired transient component in transient cavitation. We evaluated the sensitivity of the algorithm to the choice of wavelet basis using several high-order orthogonal wavelet families, including Daubechies (db10–db30), Symlets (sym10–sym20), and Coiflets (coif3–coif5). High-order wavelets in these families produced comparable separation of the periodic and transient cavitation signatures, with only minor differences in the boundaries of the reconstructed packets. Also, it was observed that increasing wavelet orders beyond a certain order, which depends on the wavelet basis functions, does not significantly improve the results. Symlet-20 wavelet was selected because it provides an appropriate compromise between time and frequency localization

while maintaining good symmetry. The depth of the decomposition was determined for each signal using the energy and kurtosis split criterion, with thresholds of 1.5 for the energy and kurtosis criteria. Nodes with energy less than 5% of the energy of the parent node are not split. The energy and kurtosis ratio thresholds are user-defined parameters controlling the sensitivity of the adaptive decomposition. For the datasets considered in this study, values between 1.5 and 3.0 produced similar decompositions, whereas lower thresholds resulted in additional subdivision of a small number of nodes. Consequently, a value of 1.5 was adopted as a practical choice that provides fine decomposition while avoiding unnecessary splitting of statistically similar subbands.

Statistical metrics defined in Section 2.2 were calculated for all leaf nodes. From the feature space analysis, a smaller subset of features—normalized spectral kurtosis, average energy, and kurtosis—was selected for clustering and classification of the packets, as detailed in Section 3.1.

To classify the leaf nodes, the algorithm was set to search between 2–5 clusters using the silhouette score. The clustering that returns the best silhouette score is ultimately selected. Clusters are sorted from highest (*Cluster 0*) to lowest average energy.

For the submicron bubble, four clusters are identified. The first cluster (*Cluster 0*) includes the periodic component, as it has the highest energy (24.9) and the lowest kurtosis (1.5) (see Fig. 4a and b). *Cluster 1* includes nodes that represent the harmonics and ultra-harmonic components, while *Cluster 2* and *Cluster 3* represent the high-frequency transient components below and above $20f_0$ respectively.

The leaf nodes of the emissions from the 3.4 micron bubble are classified into three clusters: *Cluster 0* includes the fundamental frequency, i.e., the periodic component (see Fig. 4c and d). It has the highest energy (50) and lowest kurtosis (2.0) among other cluster centers. The rest of the signal is contained within *Cluster 2* (measurement noise above 20 MHz) and *Cluster 1* (subharmonics, broadband, and second

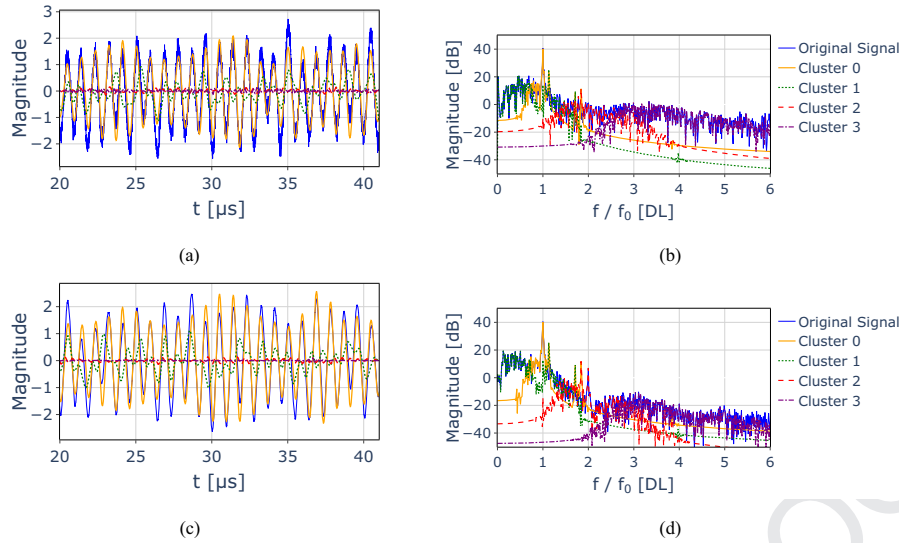


Fig. 5. Reconstructed time series of the leaf nodes of different clusters and their spectra, respectively: (a,b) D1 data, (c,d) D2 data.

harmonic). In this case, the second harmonic and broadband are not pronounced enough to be separated into different clusters.

Finally, the emissions from the large 50 micron bubble are classified into three clusters, with the fundamental in *Cluster 0*, subharmonics and second harmonic and in *Cluster 1*, and measurement noise in *Cluster 2* (see Fig. 4e and f).

3.3. Analyzing PCD data

A focused ultrasound transducer (H102, Sonic Concepts, WA, USA) was driven with a burst of 100 cycles at the fundamental frequency of 1.1 MHz and different amplitudes. A focused PCD (Olympus V392) with center frequency of 1 MHz was used to measure acoustic emissions from the focal point of the ultrasound transducer. It is mounted in the aperture of the H102 transducer; hence their focal volumes overlap. The transducers were immersed in a tank filled with de-ionized water maintained at room temperature (22 °C), with the dissolved oxygen concentration monitored and maintained at $7.00 \pm 0.2 \text{ mg L}^{-1}$ throughout all experiments. The received PCD signals were amplified, digitized at a sampling frequency of 200 MS s^{-1} , and corrected for the frequency response of the receiving transducer using time-domain deconvolution filters. The peak-positive (P_{p+}) and peak-negative (P_{p-}) acoustic pressures at the focus were estimated by numerically solving the Khokhlov–Zabolotskaya–Kuznetsov (KZK) equation using the HITU Simulator software [44], as described in Ref. [30]. A detailed description of the experimental setup, pressure estimation, calibration, data acquisition parameters, and experimental procedures is provided in Ref. [30]. Two experimental scenarios were investigated, characterized by calculated peak-positive (P_{p+}) and peak-negative (P_{p-}) pressures at the focus: i) low-amplitude sonication where $P_{p+} = 3 \text{ MPa}$ and $P_{p-} = 2.6 \text{ MPa}$, denoted as D1 below, ii) high-amplitude sonication where $P_{p+} = 12.5 \text{ MPa}$ and $P_{p-} = 7.1 \text{ MPa}$, denoted as D2. In scenario D1, the incident pressure waves remain linear, whereas in D2, the waveform becomes strongly distorted, leading to the formation of higher harmonics. These harmonics can excite bubble nuclei smaller than the resonance bubble at the excitation frequency $f = 1.1 \text{ MHz}$.

The parameters used for the algorithm are identical to those described in Section 3.2. The wavelet transform applies energy and kurtosis-based splitting using the sym20 wavelet with zero padding. It also limits the decomposition depth by requiring each subband to remain larger than $f_0/4$, ensuring every node contains sufficient data for robust feature extraction. Considering the bandwidth of the PCD, nodes with frequency content above 10 MHz are discarded.

The algorithm divides the leaf nodes of the D1 and D2 data into four clusters. In both cases, the fundamental is the high-energy node that has been extracted, denoted as *Cluster 0*, as shown in Fig. 5, where the time series reconstructed from these clusters and their spectra are presented. In both cases, the algorithm identifies a medium-kurtosis (< 3), low-energy, and high-spectral kurtosis cluster (*Cluster 2*). This cluster has characteristics of measurement noise. A high kurtosis, low-energy cluster is also identified (*Cluster 3*) and has the characteristics of broadband noise. Finally, *Cluster 1* includes packets with low spectral entropy and high energy, representing the subharmonics in the signals.

The higher intensity ultrasound produces a prominent harmonic at $2f_0$ and a sub-harmonic at $\frac{f_0}{2}$, shown in Fig. 5(d). The algorithm is able to automatically extract the fundamental frequency, harmonics, subharmonics, and broadband but does not clearly separate the broadband and measurement noise.

To characterize the broadband component, kurtosis-based thresholding is applied by selecting clusters with center kurtosis values $K > 3.2$. Subsequently, all nodes belonging to these clusters are thresholded. The time series are then reconstructed using these nodes, as shown in Fig. 6(a). Statistical features of the reconstructed broadband components are calculated, as shown in Table 1. These quantitative data show that the energy and the spectral entropy of the broadband are higher at higher intensity excitation. High kurtosis and non-zero skewness confirm the transient nature of these components. Additionally, they suggest that the broadband components are statistically non-Gaussian, especially under high-intensity excitation.

Since the broadband time series can be reconstructed, its frequency spectrum can also be computed independently. To quantify the spectral bandwidth of the broadband, we adapt established metrics from optics [45]: the root-mean square (RMS) of the spectrum, the full-width half maximum (FWHM), and the full-width $1/e^2$ of the maximum. Because the broadband spectrum is highly oscillatory, a zero-order Gaussian filter (with standard deviation equal to that of the signal) is applied before calculating the bandwidth. The results are shown in Fig. 6(b) and the calculated frequency bandwidths are listed in Table 2. The results show that the bandwidth of the spectrum decreases with excitation intensity. As indicated in Table 1, the energy of the broadband rises, suggesting that the transient impulses generated by transient cavitation at high-intensity excitation are more energetic – bubbles potentially undergo more violent collapses, consistent with the predictions from bubble dynamics.

To the best of our knowledge, the bandwidth of the broadband component in PCD data has typically been estimated heuristically, often by

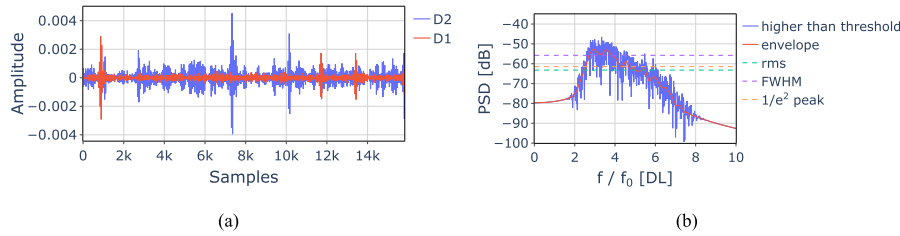


Fig. 6. (a) Reconstructed broadband components of the D1 and D2 data sets. (b) Thresholding to determine the bandwidth of the broadband component for D1 data.

Table 1

Statistical metrics of the reconstructed broadband components. Energy values are scaled back by multiplying by the standard deviation of the data from each experiment.

Experiment	Energy	Kurtosis	Skewness	Spectral entropy
D1	0.04×10^{-6}	25.5	0.11	0.58
D2	0.29×10^{-6}	12.5	0.11	0.50

Table 2

Broadband bandwidth quantified using RMS, FWHM, and $1/e^2$ criteria. Frequency ranges are normalized by the driving frequency, i.e., $f_0 = 1.1$ MHz.

Experiment	RMS [$\times f_0$]	FWHM [$\times f_0$]	$1/e^2$ [$\times f_0$]
D1	2.6 – 5.0	2.7 – 3.9	2.5 – 4.3
D2	2.4 – 4.2	2.7 – 3.7	2.5 – 4.1

visually identifying spectral elevation above the noise floor. However, the method described here enables us to extract and reconstruct the broadband time series and spectrum, and quantify its bandwidth and statistical features.

4. Discussion

In this work, we have developed a method for analyzing acoustic emissions from PCDs using DWPT and k-means clustering. The method is based on the energy- and kurtosis-split DWPT, which allows for flexible separation of transient and periodic components in the acoustic emissions. The algorithm was found to be effective in identifying components of the emissions, manifested as spectral components such as harmonics, sub-harmonics, ultra-harmonics, and broadband.

The broadband component was quantitatively characterized, as it is a strong indicator of the presence of transient cavitation. Determining a more quantitative measure of the bandwidth and energy of the broadband component will allow researchers to more precisely design instrumentation in experiments to reliably detect the presence of transient cavitation. This was achieved by thresholding for packets with kurtosis above a value of 3.2, indicating that the noise arises from a non-Gaussian distribution. The algorithm extracts broadband noise with a bandwidth around $2.5f_0 - 4f_0$ in both sets of PCD data and calculates an energy about 7 times larger for the higher intensity ultrasound. The algorithm is also able to extract sharp transients from simulated bubble dynamics data as part of the broadband component, arising from inertial collapses of bubbles.

More generally, the proposed framework separates, reconstructs, and quantitatively characterises the individual components of acoustic emissions. Unlike conventional spectral and time-frequency methods which primarily provide global spectral or time-frequency descriptors, the proposed method isolates statistically distinct components, including the fundamental, harmonics, sub-harmonics, ultra-harmonics, broadband component, and measurement noise. This enables each component to be analysed independently in terms of its energy, bandwidth,

and statistical properties, providing a more comprehensive quantitative characterization of cavitation emissions.

The k-means algorithm worked efficiently and consistently for clustering, often converging within 5 iterations onto a solution. However, it was sensitive to changes in the energy threshold, E_t , in the decomposition algorithm (see Section 2.1). This occurs because the data is sparse and contains non-spherical clusters which violates the assumptions of the k-means algorithm. More flexible clustering methods, such as Gaussian mixture models, could be explored to improve reliability by accommodating clusters of different shapes.

5. Conclusion

This paper presents a novel approach for analyzing PCDs-measured acoustic emissions, utilizing multi-resolution analysis through DWPT and k-means clustering. The combination of DWPT with a node selection algorithm based on splitting the signal to find high energy and kurtosis components proved effective in flexibly separating the transient and periodic components of the emissions. Several metrics were evaluated to classify the leaf nodes and quantitatively characterize cavitation features. SOMs were used to visualize the feature space and identify a subset of the most relevant metrics. SOMs calculated using Gilmore simulations showed that the non-redundant descriptors are average energy, kurtosis, and normalized spectral kurtosis.

The algorithm was validated using both numerical experiments and experimental measurements, successfully distinguishing the periodic and transient components of the emissions. It also identified and characterized cavitation features such as harmonics, sub-harmonics, ultra-harmonics, and broadband components. This was achieved by linking the physical interpretation of these metrics to the time-frequency signatures of acoustic emissions. A quantitative approach was proposed and utilized to determine the bandwidth and statistical metrics of the broadband component. This provides an effective tool for quantifying various cavitation features in PCD data and for quantitatively characterizing acoustic cavitation.

Since the statistical characteristics of the measurement noise are unknown, it is challenging to separate the measurement noise from broadband component of transient cavitation, particularly in low-kurtosis cases. In addition, misclassifications may occur when DWPT nodes lie at the boundaries of clusters or when the clustering structure is not well-separated. These issues stem from inherent limitations of k-means, such as its reliance on Euclidean distance, assumption of equally sized clusters, and hard assignments. Future works will explore alternative classification and decomposition methods to improve robustness in these cases.

CRedit authorship contribution statement

Max Au-Yeung: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Reza Haqshenas:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Reza Haqshenas reports that financial support was provided by Focused Ultrasound Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data and code are openly available on GitHub page.

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